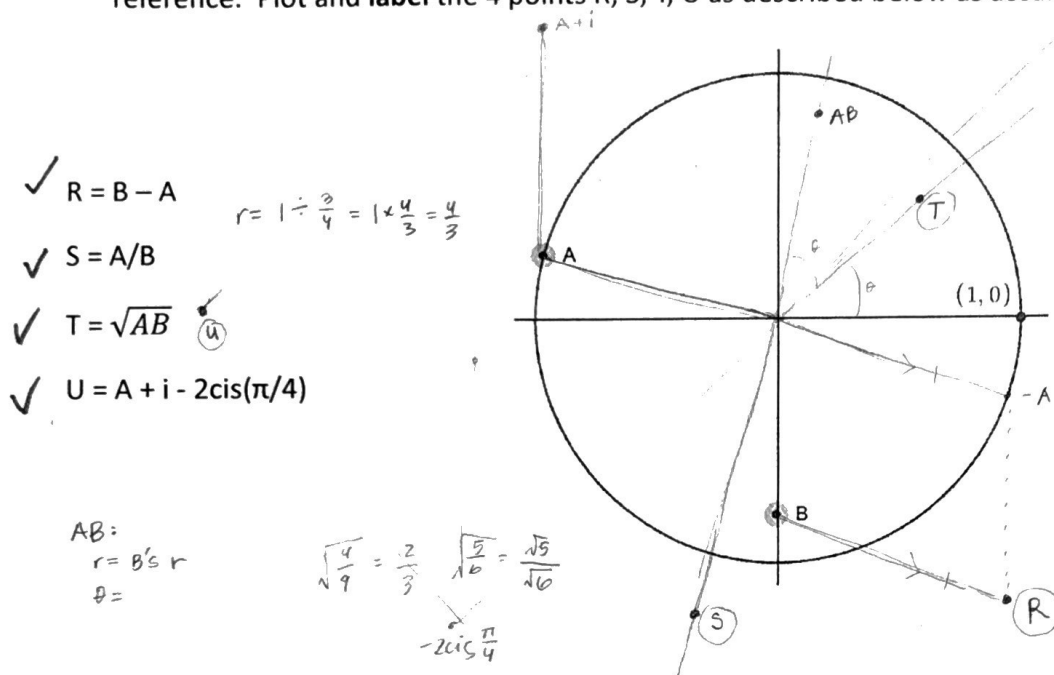
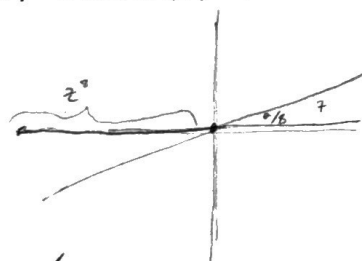


1. Complex numbers A and B are graphed below on the real-imaginary plane. A unit circle is given for reference. Plot and label the 4 points R, S, T, U as described below as accurately as possible: [6 pts]



2. Find 2 complex numbers z in the first quadrant (in cis form) such that..... [4]

$|z| = 7$ and $\text{Re}(z^8) < 0$ and $\text{Im}(z^8) = 0$



θ of z^8 is π ,

so θ of z is $\frac{\pi}{8}$

given that r of $z = 7$

or -7

$\pi + \frac{\pi}{8}$

$\frac{8\pi}{8} + \frac{\pi}{8}$

$z_1 = 7 \text{cis} \frac{\pi}{8}$ ✓

$z_2 = -7 \text{cis} \frac{9\pi}{8} = 7 \text{cis} \frac{3\pi}{8}$

3. A decagon is a polygon with 10 sides. [2 pts each]

- a) Write a single 2×2 matrix that would generate a group

isomorphic to the cyclic group C_{10} .

$\begin{bmatrix} \cos 36^\circ & -\sin 36^\circ \\ \sin 36^\circ & \cos 36^\circ \end{bmatrix}$

$\begin{bmatrix} \cos 36^\circ & -\sin 36^\circ \\ \sin 36^\circ & \cos 36^\circ \end{bmatrix}$ ✓

- b) Write a single complex number in (cis form) that would generate

a group isomorphic to the cyclic group C_{10}

$\text{cis } 36^\circ$ ✓

$\cos 36^\circ + i \sin 36^\circ$



$$\frac{0.60}{6 \mid 360}$$

$$\cos 60 = \frac{1}{2}$$

$$\sin 60 = \frac{\sqrt{3}}{2}$$



4a) Write 2 matrices that would, together, generate D_6 , the dihedral (reflection/rotation) group of a hexagon, or state that this is not possible. [2]

order = 12

$$\begin{bmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix}$$

rot. by 60°

$$\text{and } \begin{bmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix}$$

ref. over $y = (\tan 60^\circ)x$

rotation by 60° and ref. over $y = \tan 60^\circ x$

$$\begin{bmatrix} \cos 120 & \sin 120 \\ \sin 120 & -\cos 120 \end{bmatrix}$$

4b) Write 2 complex numbers that would, together, generate D_6 , the dihedral (reflection) group of a hexagon, or state that this is not possible. [2]

not possible

5. Is the conjugate of the product of 2 complex numbers equal to the product of their individual conjugates? Provide evidence that your answer is correct. [3]

$$a = 2+3i \quad b = 5+4i$$

$$ab : (2+3i)(5+4i) = 10 + 8i + 15i - 12 = -2 + 23i \rightarrow -2 - 23i$$

$$\bar{a}\bar{b} : (2-3i)(5-4i) = 10 - 8i - 15i - 12 = -2 - 23i$$

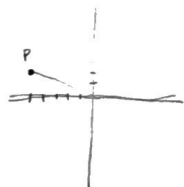
equal

YES

You did this for 2 specific complex number not for any $a+bi, c+di$

6. Consider the point $P(-5,2)$ in the 2nd quadrant. [2 each]

a) Write a matrix that would rotate P around the origin so that its image was on the negative x axis.



$$\tan \theta = \frac{2}{5}$$

$$\theta = \tan^{-1}\left(\frac{2}{5}\right)$$

$$\begin{bmatrix} \cos(\tan^{-1}(\frac{2}{5})) & -\sin(\tan^{-1}(\frac{2}{5})) \\ \sin(\tan^{-1}(\frac{2}{5})) & \cos(\tan^{-1}(\frac{2}{5})) \end{bmatrix}$$

✓

b) Write a matrix that would shear P so that its image is on the negative x-axis.

$$\frac{2}{5} \times 5$$

$$x + Ky =$$

$$\begin{bmatrix} 1 & 0 \\ K & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ y + Kx \end{bmatrix}$$

$$y + Kx = 0$$

$$2 + K(-5) = 0$$

$$-5K = -2$$

$$K = \frac{2}{5}$$

$$\begin{bmatrix} 1 & 0 \\ \frac{2}{5} & 1 \end{bmatrix}$$

✓

$$\begin{bmatrix} 1 & 0 \\ \frac{2}{5} & 1 \end{bmatrix} \begin{bmatrix} -5 \\ 2 \end{bmatrix} = \begin{bmatrix} -5 \\ 0 \end{bmatrix}$$

c) Write a matrix (not a shear) that would map EVERY point (not just P) in the plane to the line $y=5x$.

$$\begin{bmatrix} 1 & 1 \\ 5 & 5 \end{bmatrix}$$

✓

$$\begin{bmatrix} 1 & 1 \\ 5 & 5 \end{bmatrix} \begin{bmatrix} -5 \\ 2 \end{bmatrix} = \begin{bmatrix} -5 \\ -5 \end{bmatrix}$$

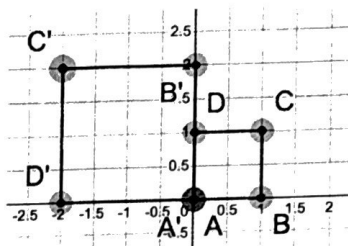
Below are 4 different images of the unit square under a transformation. In each case write a single matrix that would achieve the transformation. Then describe the transformation(s) in words using the correct order when needed. Include specifics, i.e. don't just say "shear" say the direction and magnitude as well.

[3 pts each]

a) Matrix: $\begin{bmatrix} 0 & -2 \\ 2 & 0 \end{bmatrix}$ ✓

$$\begin{bmatrix} 0 & -2 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -2 \\ 2 \end{bmatrix}$$

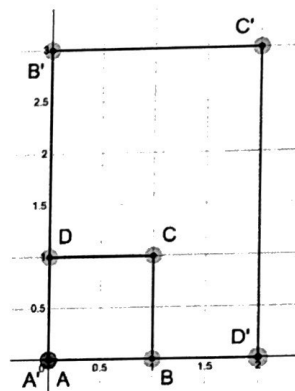
$$\begin{bmatrix} 0 & -2 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -2 \\ 2 \end{bmatrix}$$



Words: dilation of factor 2, rotation 90° CCW ✓

b) Matrix: $\begin{bmatrix} 0 & 2 \\ 3 & 0 \end{bmatrix}$ ✓

$$\begin{bmatrix} 0 & 2 \\ 3 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

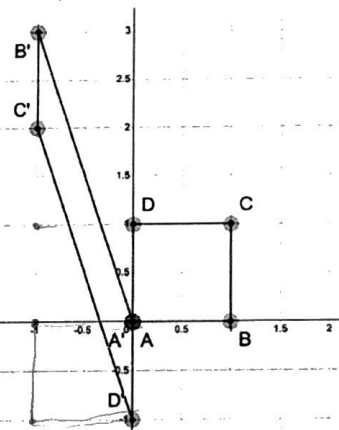


Words: reflect over $y=x$, stretch by 2 in $(+)$ x, stretch by 3 in $(+)$ y direction ✓

c) Matrix: $\begin{bmatrix} -1 & 0 \\ 3 & -1 \end{bmatrix}$ ✓

$$\begin{bmatrix} -1 & 0 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 0 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$



Words: rotation of 180°, then shear by $f=3$ in positive y direction ✓

$$\begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$$

d) Matrix: $\begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$ ✓

Words: translation right 3, up 2 ✓

